Stat 110 Strategic Practice 10 & Penultimate Homework: Solutions

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Strategic Practice 10 Solutions

1. Conditional Expectation and Conditional Variance

Problem 1. We verify that $E((Y - E(Y|X))^2 \mid X) = E(Y^2 \mid X) - (E(Y|X))^2$.

Write $g(X) = E(Y|X)$. Expanding the square inside the conditional expectation:

$$E((Y - g(X))^2 \mid X) = E(Y^2 - 2Yg(X) + g(X)^2 \mid X).$$

By linearity of conditional expectation and the rule of taking out what is known (since $g(X)$ is a function of X and thus fixed when we condition on X):

$$= E(Y^2 \mid X) - 2g(X)E(Y \mid X) + g(X)^2 = E(Y^2 \mid X) - 2g(X)^2 + g(X)^2 = E(Y^2 \mid X) - g(X)^2.$$

Problem 2. (Eve's Law) We prove $\mathrm{Var}(Y) = E(\mathrm{Var}(Y \mid X)) + \mathrm{Var}(E(Y \mid X))$.

Let $g(X) = E(Y \mid X)$. By Adam's Law, $E(g(X)) = E(Y)$.

From the definition $\mathrm{Var}(Y \mid X) = E(Y^2 \mid X) - g(X)^2$, taking the unconditional expectation:

$$E(\mathrm{Var}(Y \mid X)) = E(Y^2) - E(g(X)^2).$$

For the second term, since $E(g(X)) = E(Y)$:

$$\mathrm{Var}(E(Y \mid X)) = E(g(X)^2) - (E(g(X)))^2 = E(g(X)^2) - (E(Y))^2.$$

Adding the two expressions, $E(g(X)^2)$ cancels and we obtain

$$E(\mathrm{Var}(Y \mid X)) + \mathrm{Var}(E(Y \mid X)) = E(Y^2) - (E(Y))^2 = \mathrm{Var}(Y).$$

Problem 3. $W = Y - E(Y \mid X)$ is the prediction residual.

(a) By Adam's Law and linearity,

$$E(W) = E(Y) - E(E(Y \mid X)) = E(Y) - E(Y) = 0.$$

Conditioning on X and taking out what is known ($E(Y \mid X)$ is a function of X):

$$E(W \mid X) = E(Y \mid X) - E(E(Y \mid X) \mid X) = E(Y \mid X) - E(Y \mid X) = 0.$$

(b) With $W \mid X \sim N(0, X^2)$ and $X \sim N(0, 1)$, Eve's Law gives

$$\mathrm{Var}(W) = E(\mathrm{Var}(W \mid X)) + \mathrm{Var}(E(W \mid X)) = E(X^2) + \mathrm{Var}(0) = E(X^2) = 1.$$

Problem 4. Emails arrive with inter-arrival times i.i.d. Expo(λ); each is non-spam with probability p.

Let N be the index of the first non-spam email, so $N - 1 \sim \mathrm{Geom}(p)$ (number of spam emails before the first non-spam). Write $X = X_1 + X_2 + \cdots + X_N$ where X_j is the inter-arrival time before the jth email, each Expo(λ) and independent.

(a) By Adam's Law, $E(X \mid N = n) = n/\lambda$, so

$$E(X) = E(N/\lambda) = \frac{E(N)}{\lambda} = \frac{1}{\lambda p}.$$

For the variance, by Eve's Law:

$$\mathrm{Var}(X) = E(\mathrm{Var}(X \mid N)) + \mathrm{Var}(E(X \mid N)) = E\left(\frac{N}{\lambda^2}\right) + \mathrm{Var}\left(\frac{N}{\lambda}\right) = \frac{E(N)}{\lambda^2} + \frac{\mathrm{Var}(N)}{\lambda^2}.$$

With $N \sim \mathrm{Geom}(p)$ (support $\{1, 2, \dots\}$): $E(N) = 1/p$ and $\mathrm{Var}(N) = q/p^2$. So

$$\mathrm{Var}(X) = \frac{1}{\lambda^2 p} + \frac{q}{\lambda^2 p^2} = \frac{1}{\lambda^2 p^2}.$$

(b) Conditioning on N and using the independence of the X_j's:

$$E(e^{tX}) = E\big(E(e^{tX} \mid N)\big) = E\big(M_1(t)^N\big),$$

where $M_1(t) = \frac{\lambda}{\lambda - t}$ is the MGF of a single $\mathrm{Expo}(\lambda)$. By LOTUS for the Geometric:

$$E(M_1(t)^N) = \sum_{n=1}^{\infty} M_1(t)^n \cdot p\, q^{n-1} = \frac{p}{q} \sum_{n=1}^{\infty} (q\, M_1(t))^n = \frac{p}{q} \cdot \frac{q\, M_1(t)}{1 - q\, M_1(t)} = \frac{p M_1(t)}{1 - q M_1(t)}.$$

Substituting $M_1(t) = \lambda/(\lambda - t)$:

$$\frac{p\lambda/(\lambda - t)}{1 - q\lambda/(\lambda - t)} = \frac{p\lambda}{(\lambda - t) - q\lambda} = \frac{p\lambda}{p\lambda - t} = \frac{\lambda p}{\lambda p - t}.$$

This is the $\mathrm{Expo}(\lambda p)$ MGF, so $X \sim \mathrm{Expo}(\lambda p)$.

Problem 5. A coin is picked at random from $\{p_1, p_2\}$; it is flipped n times and X counts Heads. Let I be the indicator of picking coin p_1. By Adam's Law,

$$E(X) = E(E(X \mid I)) = E(I n p_1 + (1 - I) n p_2) = \frac{n(p_1 + p_2)}{2}.$$

By Eve's Law, $\mathrm{Var}(X \mid I) = I\, n p_1(1 - p_1) + (1 - I) n p_2(1 - p_2)$, which equals $np_1(1 - p_1)$ or $np_2(1 - p_2)$ each with probability $1/2$. Also $E(X \mid I)$ takes values np_1 or np_2 each with probability $1/2$. So

$$E(\mathrm{Var}(X \mid I)) = \frac{n}{2}(p_1(1 - p_1) + p_2(1 - p_2)),$$

$$\mathrm{Var}(E(X \mid I)) = \frac{n^2}{4}(p_1 - p_2)^2.$$

Therefore,

$$\mathrm{Var}(X) = \frac{n}{2}(p_1(1 - p_1) + p_2(1 - p_2)) + \frac{n^2}{4}(p_1 - p_2)^2.$$

2. Inequalities

For all parts, X and Y are i.i.d. positive r.v.s, $c > 0$.

(a) $E(\ln X) \le \ln(E(X))$: the log function is concave, so Jensen gives $\le$.

(b) $E(X) \le \sqrt{E(X^2)}$: since $\mathrm{Var}(X) \ge 0$, we have $(E(X))^2 \le E(X^2)$, so take square roots.

(c) $E(\sin^2 X) + E(\cos^2 X) = 1$: by linearity and the identity $\sin^2 + \cos^2 = 1$.

(d) $E(|X|) \le \sqrt{E(X^2)}$: since $X > 0$, $|X| = X$, so this is the same as (b).

(e) $P(X > c) \le E(X^3)/c^3$: applying Markov's inequality to X^3 with threshold c^3.

(f) $P(X \le Y) = P(X \ge Y)$: wait, by symmetry (i.i.d.) $P(X \le Y) = P(Y \le X) = P(X \ge Y)$, so $=$.

(g) $E(XY) \le \sqrt{E(X^2)E(Y^2)}$: by the Cauchy-Schwarz inequality.

(h) $P(X + Y > 10) \le P(X > 5 \text{ or } Y > 5)$: if $X + Y > 10$ then $\max(X, Y) > 5$, i.e., $X > 5$ or $Y > 5$.

(i) $E(\min(X, Y)) \le \min(E(X), E(Y))$: $\min(X, Y) \le X$ pointwise, so take expectation; similarly for Y.

(j) $E(X/Y) \ge E(X)/E(Y)$: since $E(1/Y) \ge 1/E(Y)$ by Jensen (as $1/y$ is convex), and $E(X/Y) = E(X)E(1/Y)$ by independence.

(k) $E(X^2(X^2 + 1)) \ge E(X^2(Y^2 + 1))$: both sides equal $E(X^4) + E(X^2)$. Since X, Y are i.i.d., $E(X^4)$ appears on the left and $E(X^2Y^2) = E(X^2)E(Y^2) = (E(X^2))^2 \le E(X^4)$ on the right. So $\ge$.

(l) $E(X^3/(X^3 + Y^3)) = E(Y^3/(X^3 + Y^3))$: the two expressions sum to 1 and are equal by symmetry (i.i.d.), so each equals 1/2.

Problem 2(a). $g(x) = 1/x$ is strictly convex on $(0, \infty)$ since $g''(x) = 2x^{-3} > 0$. By Jensen's inequality (strict for non-constant r.v.s),

$$E(1/X) > 1/E(X).$$

Problem 2(b). Let $W = Y/X$, a positive non-constant r.v. By part (a),

$$E(X/Y) = E(1/W) > \frac{1}{E(W)} = \frac{1}{E(Y/X)},$$

so $E(X/Y) \cdot E(Y/X) > 1$.

Problem 3. We want $P(|\bar{X}_n - \mu| > 2\sigma) \le 0.01$. By Chebyshev's inequality,

$$P(|\bar{X}_n - \mu| > 2\sigma) \le \frac{\mathrm{Var}(\bar{X}_n)}{(2\sigma)^2} = \frac{\sigma^2/n}{4\sigma^2} = \frac{1}{4n}.$$

Setting $1/(4n) \le 1/100$ gives $n \ge 25$.

Problem 4. (AM-GM via Jensen) Assume the a_j are distinct (if not, assign each distinct value a probability proportional to its multiplicity). Let X take value a_j with probability $1/n$ for each j. Since log is concave, Jensen gives $E(\log X) \le \log(E(X))$:

$$\frac{1}{n}\sum_{j=1}^{n} \log a_j \le \log\left(\frac{a_1 + \cdots + a_n}{n}\right).$$

Exponentiating both sides and using $\exp\left(\frac{1}{n}\sum \log a_j\right) = (a_1 \cdots a_n)^{1/n}$:

$$(a_1 a_2 \cdots a_n)^{1/n} \le \frac{a_1 + a_2 + \cdots + a_n}{n}.$$

Penultimate Homework Solutions

1. Judit's Tournament Wins

$N \sim \text{Geom}(s)$ (number of tournaments, support $\{0, 1, 2, \dots\}$ here treated as $\{1, 2, \dots\}$; we take $N \sim \text{Geom}(s)$ with $E(N) = (1-s)/s$), and $T \mid N \sim \text{Bin}(N, p)$.

(a) By Adam's Law:

$$E(T) = E(E(T \mid N)) = E(Np) = p \cdot \frac{1-s}{s}.$$

By Eve's Law:

$$\text{Var}(T) = E(\text{Var}(T \mid N)) + \text{Var}(E(T \mid N)) = E(Np(1-p)) + \text{Var}(Np) = p(1-p)\frac{1-s}{s} + p^2\frac{1-s}{s^2}.$$

Combining over a common denominator,

$$\text{Var}(T) = \frac{p(1-s)(s+p(1-s))}{s^2}.$$

(b) The MGF of T is found by conditioning on N. Each tournament win indicator is Bernoulli(p) independently, so

$$E(e^{tT}) = E\big(E(e^{tT} \mid N)\big) = E\big((pe^t+q)^N\big) = \sum_{n=0}^{\infty}(pe^t+q)^n s(1-s)^n = \frac{s}{1-(1-s)(pe^t+q)}.$$

This matches the MGF of a Geom(θ) distribution (for an appropriate θ). Setting $\theta = s/(s+p(1-s))$ puts it in standard form, showing $T \sim \text{Geom}\Big(\frac{s}{s+p(1-s)}\Big)$.

Story proof: Each event in Judit's career is a win (probability $(1-s)p$), a loss (probability $(1-s)(1-p)$), or retirement (probability s). Conditioning on the outcome being either a win or retirement (i.e., ignoring losses), the probability that a non-loss outcome is a retirement is $\frac{s}{s+p(1-s)}$, and T counts the wins before the first retirement. This is Geometric with the stated parameter.

2. Conditional Expectation of a Weighted Sum

Since X_1 and X_2 are i.i.d., by symmetry $E(X_1 \mid \bar{X}) = E(X_2 \mid \bar{X})$. By linearity and taking out what is known,

$$E(X_1 \mid \bar{X}) + E(X_2 \mid \bar{X}) = E(X_1 + X_2 \mid \bar{X}) = 2\bar{X},$$

so $E(X_1 \mid \bar{X}) = E(X_2 \mid \bar{X}) = \bar{X}$. Therefore, for any constants w_1, w_2 with $w_1 + w_2 = 1$,

$$E(w_1X_1 + w_2X_2 \mid \bar{X}) = w_1\bar{X} + w_2\bar{X} = \bar{X}.$$

3. Stock Volatility: Mixture of Normals

$X = I_1X_1 + I_2X_2$ with $I_1 \sim \text{Bern}(p)$, $I_2 = 1 - I_1$, $X_j \sim N(0, \sigma_j^2)$ all independent.

(a) *Via Eve's Law:* Given I_1, X equals $X_1 \sim N(0, \sigma_1^2)$ or $X_2 \sim N(0, \sigma_2^2)$, so $E(X \mid I_1) = 0$ always and $\text{Var}(X \mid I_1) = I_1\sigma_1^2 + (1-I_1)\sigma_2^2$. By Eve's Law,

$$\text{Var}(X) = E(I_1\sigma_1^2 + (1-I_1)\sigma_2^2) + \text{Var}(0) = p\sigma_1^2 + (1-p)\sigma_2^2.$$

Direct covariance method: Since $I_1I_2 = 0$ always, all cross-terms $\text{Cov}(I_1X_1, I_2X_2) = E(I_1I_2X_1X_2) = 0$. Also $E(I_1X_1) = 0$ (as $E(X_1) = 0$), so

$$\text{Var}(I_1X_1) = E(I_1^2X_1^2) = E(I_1)E(X_1^2) = p\sigma_1^2.$$

Similarly $\text{Var}(I_2X_2) = q\sigma_2^2$. So $\text{Var}(X) = p\sigma_1^2 + q\sigma_2^2$.

(b) Since $I_1 I_2 = 0$ and $I_j^k = I_j$ for any $k \geq 1$:

$$X^4 = (I_1 X_1 + I_2 X_2)^4 = I_1 X_1^4 + I_2 X_2^4.$$

For $Z \sim N(0, \sigma^2)$, $E(Z^4) = 3\sigma^4$. So

$$E(X^4) = E(I_1)E(X_1^4) + E(I_2)E(X_2^4) = 3p\sigma_1^4 + 3q\sigma_2^4.$$

With $\mu = E(X) = 0$ and $\sigma_X^2 = p\sigma_1^2 + q\sigma_2^2$,

$$\text{Kurt}(X) = \frac{E(X^4)}{\sigma_X^4} - 3 = \frac{3(p\sigma_1^4 + q\sigma_2^4)}{(p\sigma_1^2 + q\sigma_2^2)^2} - 3.$$

When $\sigma_1 = \sigma_2$, this reduces to $3\sigma^4/\sigma^4 - 3 = 0$ (Gaussian kurtosis). When $\sigma_1 \neq \sigma_2$, the numerator exceeds the denominator by Jensen's inequality applied to the convex function t^2: $E(Y^2) > (E(Y))^2$ for the r.v. Y taking values σ_1^2 or σ_2^2. So the kurtosis is strictly positive, reflecting heavier tails than a Normal.

4. Unbiased vs. Bayes Estimator

(a) Let $\hat{\theta}$ be unbiased, so $E(\hat{\theta} \mid \theta) = \theta$. Conditioning on θ:

$$E(\hat{\theta}-\theta)^2 = E\Big[E\Big((\hat{\theta} - \theta)^2 \mid \theta\Big)\Big] = E\Big[E(\hat{\theta}^2 \mid \theta) - 2\theta E(\hat{\theta} \mid \theta) + \theta^2\Big] = E(\hat{\theta}^2) - 2E(\theta^2) + E(\theta^2) = E(\hat{\theta}^2) - E(\theta^2).$$

(b) Now $\hat{\theta} = E(\theta \mid X)$ (the Bayes procedure). Conditioning on X (and using the fact that $\hat{\theta}$ is a function of X, hence known given X):

$$E(\hat{\theta}-\theta)^2 = E\Big[E\Big((\hat{\theta} - \theta)^2 \mid X\Big)\Big] = E\Big[\hat{\theta}^2 - 2\hat{\theta}E(\theta \mid X) + E(\theta^2 \mid X)\Big] = E(\hat{\theta}^2) - 2E(\hat{\theta}^2) + E(\theta^2) = E(\theta^2) - E(\hat{\theta}^2).$$

(c) If $\hat{\theta}$ were both Bayes and unbiased, then from (a) the MSE is $E(\hat{\theta}^2) - E(\theta^2)$ and from (b) it is $E(\theta^2) - E(\hat{\theta}^2)$. Both must be equal to the same non-negative quantity, so $E(\hat{\theta}^2) - E(\theta^2) = E(\theta^2) - E(\hat{\theta}^2)$, giving $2E(\hat{\theta}^2) = 2E(\theta^2)$. Combined with $E(\hat{\theta} - \theta)^2 = 0$, the non-negative r.v. $(\hat{\theta} - \theta)^2$ has mean zero, forcing $P(\hat{\theta} = \theta) = 1$. This means the data X reveal θ perfectly, which is only possible in degenerate cases where there is no statistical uncertainty left after observing X.

5. Entropy

(a) Since $x \mapsto x^3$ is a one-to-one function, X^3 takes distinct values a_j^3 each with the same probability p_j as X takes value a_j. The entropy only depends on the probability distribution, not the labels, so $H(X^3) = H(X)$.

For X^2: let X take values -1 and $+1$ each with probability $1/2$. Then $X^2 = 1$ always, so $H(X^2) = 0$, whereas $H(X) = \log_2 2 = 1$. This works because $x \mapsto x^2$ is not one-to-one.

(b) Let Y be a r.v. taking value p_j with probability p_j for each j. Then $H(X) = E(\log_2(1/Y))$. Also,

$$E(1/Y) = \sum_{j=1}^{n} p_j \cdot \frac{1}{p_j} = n.$$

Since $\log_2$ is concave, Jensen's inequality gives

$$H(X) = E(\log_2(1/Y)) \leq \log_2(E(1/Y)) = \log_2 n.$$

Equality holds if and only if $1/Y$ is constant, i.e., all p_j are equal to $1/n$. So the entropy is maximized by the uniform distribution, which is the intuitively natural result: maximum uncertainty about X's value corresponds to maximum information gained upon observing it.

6. Survey Sampling Bound

Let X_i be the indicator that person i in the sample supports the policy, so $X_i \overset{iid}{\sim} \text{Bern}(p)$ and $\hat{p} = \bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then $E(\hat{p}) = p$ and $\text{Var}(\hat{p}) = p(1-p)/n$.

By Chebyshev's inequality,

$$P(|\hat{p} - p| > c) \leq \frac{\text{Var}(\hat{p})}{c^2} = \frac{p(1-p)}{nc^2}.$$

Since $p(1-p) \leq 1/4$ for all $p \in [0,1]$ (maximized at $p = 1/2$),

$$P(|\hat{p} - p| > c) \leq \frac{1}{4nc^2}.$$